

# Polynomial Regression: L1 and L2 Comparison

## IMT2024031 | Revised submission

### 1. Objective and method

Predict the continuous target  $y$  for two personalised datasets using multivariate polynomial regression. The allowed maximum total degrees are 10 for var1 and 20 for var2. Each training and test dataset has 1,000 rows; var1 has six inputs and var2 has three.

PolynomialFeatures generates non-constant monomials with total degree at most  $d$ . StandardScaler is fitted on each training fold only. Lasso adds an L1 penalty to encourage sparse coefficients; Ridge adds an L2 penalty to shrink coefficients. Both retain an unpenalized intercept. Elastic Net is not used.

### 2. Submitted configurations and verified results

| Problem | Model      | Degree | Alpha      | OOF MSE  | OOF R2   |
|---------|------------|--------|------------|----------|----------|
| var1    | Lasso (L1) | 5      | 0.00788046 | 0.327877 | 0.965743 |
| var2    | Lasso (L1) | 11     | 0.00045000 | 0.235043 | 0.994675 |

Five shuffled folds with `random_state=42` generate out-of-fold (OOF) predictions. MSE and R2 above are computed over the pooled OOF predictions. The final model is then refitted on all training rows before generating test predictions.

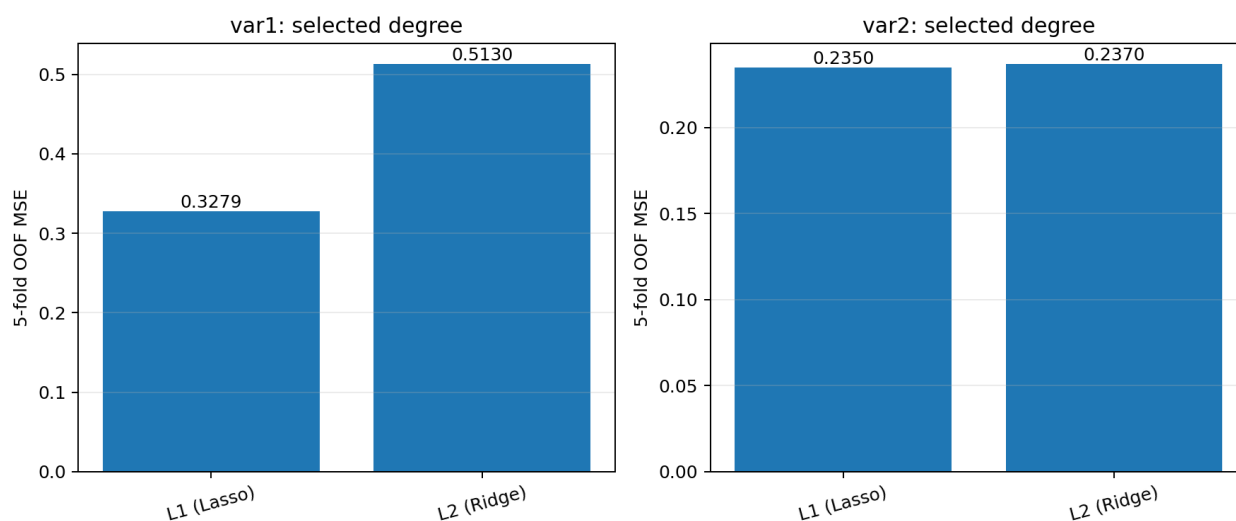
**Validation limitation.** These folds also informed model selection. Therefore, these are tuning-CV scores, not independent estimates from nested CV or a held-out test set. OOF predictions exclude each row from its fold training, but hyperparameter selection still uses the overall CV results. Test targets are hidden; no test accuracy is claimed. R2 is a variance-explained measure, not classification accuracy.

var1 retains 149 of 461 non-constant coefficients; var2 retains 163 of 363. The intercept is additional. Independent refits reproduce both submitted prediction files to numerical precision.

### 3. L1 versus L2

| Problem | Degree | Lasso MSE | Ridge MSE | Reduction vs Ridge |
|---------|--------|-----------|-----------|--------------------|
| var1    | 5      | 0.327877  | 0.513036  | 36.1%              |
| var2    | 11     | 0.235043  | 0.236987  | 0.8%               |

Ridge alpha values are 28.072162 for var1 and 1.610262 for var2. A separate sweep across all allowed degrees and the supplied Ridge alpha grid confirms these are its best configurations within that grid. Lasso has lower observed tuning-CV MSE for both datasets; the small var2 difference does not establish a statistically reliable advantage.



### 4. Search evidence and reproducibility

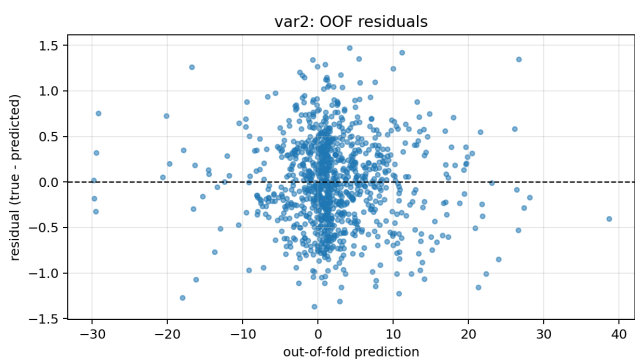
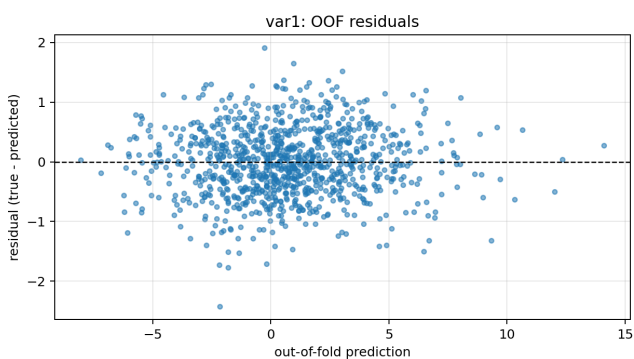
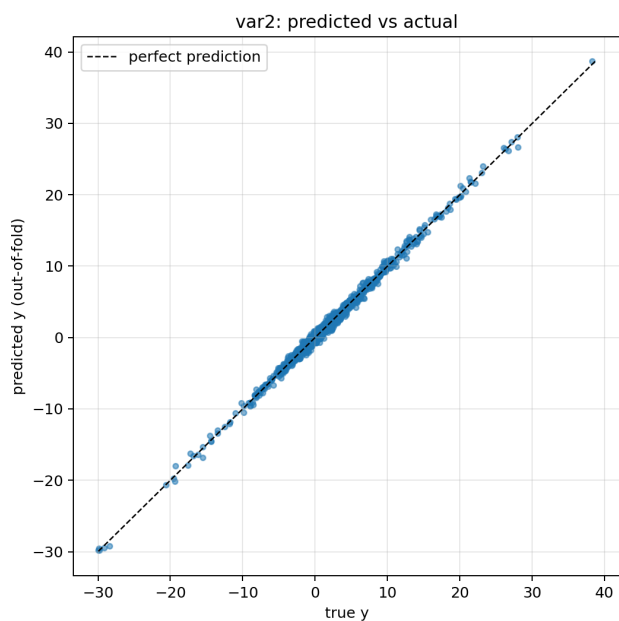
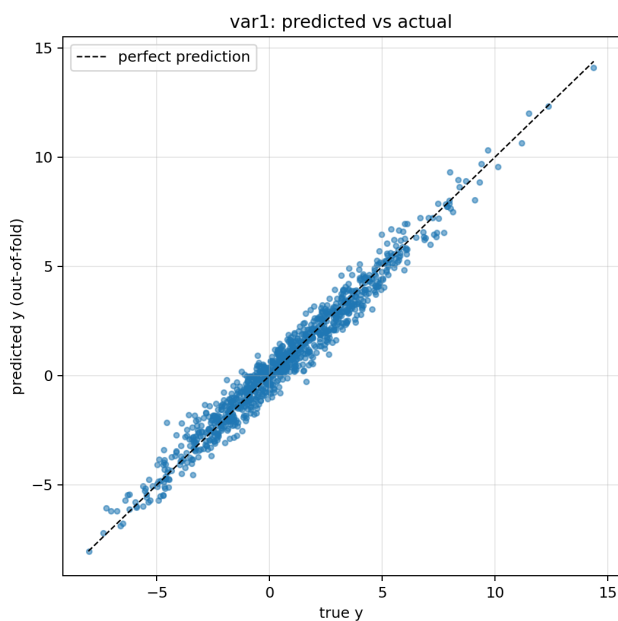
The original package contained selected-degree comparison values but no complete search tables. This revision adds completed independent search audit results and a script covering degrees 1-10 and 1-20, using the original alpha grids. The run was stopped for the submission deadline; results/search\_coverage.csv explicitly marks completed and pending degrees. The audit uses warm-start Lasso paths (10,000 iterations, or 2,000 for var1 degrees 7-10 and var2 degrees 15-20; tolerance 1e-4), and an SVD Ridge solution. For completed degrees, it records every alpha, CV MSE, and convergence diagnostics; it is not a reconstruction of missing historical logs.

The original cold-start selection script is also retained (maximum 100,000 Lasso iterations). Warm starts and stopping tolerances can cause small differences in Lasso results. Non-converged audit candidates are explicitly flagged and must not be treated as certified minima. The submitted configurations and predictions are preserved; the report does not claim they are globally optimal over every possible search setting.

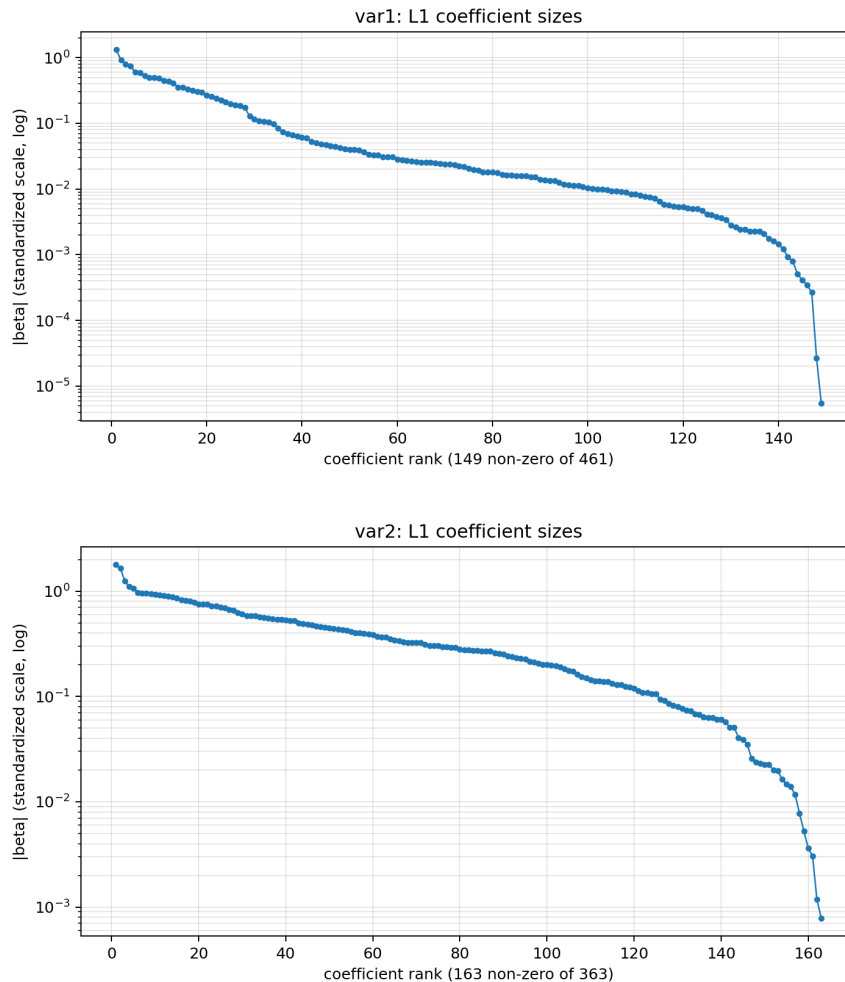
See results/search\_audit\_all\_candidates.csv and the two per-degree audit summaries. To reproduce the audit, run `python3 code/search_audit.py`. To reproduce the original cold-start search, run `python3 code/model_selection_l1_l2.py`. The latter can take substantially longer.

## 5. Out-of-fold diagnostics

Predicted-versus-actual points cluster near the equality line. Residuals are broadly centered around zero; these visual diagnostics do not prove independence or constant variance.



## 6. Sparsity and implementation



The coefficient plots show sorted nonzero magnitudes on a log scale. Their labels report the nonzero count and full feature count. Coefficients refer to standardized polynomial features, so they should not be read as raw-unit effects.

## 7. Reproduction and deliverables

Install dependencies with `python3 -m pip install -r requirements.txt`. Then run:

```
python3 code/train_final.py
python3 code/plots.py
python3 code/verify_submission.py
```

Verification refits both models and regenerates OOF predictions in memory, checks the saved predictions and residuals within numerical tolerance, and writes a verification summary. It does not overwrite the submitted prediction files. Plot summaries read metrics from CSVs rather than hardcoded values.

Submit this four-page PDF, the two predictions/IMT2024031\_pred\_var\*.csv files, and the GitHub repository containing code and supporting evidence. Each prediction CSV contains only  $y$ , with 1,000 rows in original test order.

GitHub: [https://github.com/Heer-create-lgtm/ml\\_assignment\\_1](https://github.com/Heer-create-lgtm/ml_assignment_1)